

# Group 16: Identifying and Minimizing Sources of Uncertainty in Chest X-Ray Classification Models

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**Abstract**—Chest X-rays (CXRs) are critical for diagnosing pulmonary conditions but are prone to interpretation challenges due to image variability and labeling inconsistencies. Deep learning models such as ResNet50 and DenseNet121 improve classification but struggle with uncertainty, leading to overconfident errors. This work explores whether preprocessing techniques, specifically unsharp masking and histogram equalization, can enhance classification reliability. Using the CheXpert dataset, we compare raw and preprocessed images at three resolutions: 224×224, 384×384, and 512×512. We incorporate Grad-CAM to analyze uncertainty sources and assess how preprocessing and resolution scaling influence model interpretability. Our results indicate that DenseNet121 at 224×224 resolution achieved the best balance between classification accuracy, computational efficiency, and interpretability, although uncertainty quantification remained challenging, particularly for Atelectasis and Edema.

**Index Terms**—Chest X-ray classification, uncertainty quantification, deep learning, ResNet50, DenseNet121, Grad-CAM, image preprocessing, medical AI, CheXpert dataset

## I. INTRODUCTION

Interpretation of chest X-rays (CXRs) is critical for the diagnosis of pulmonary and cardiovascular diseases, but challenges arise due to image variability, subtle overlap of features, and inconsistencies in labeling. Even experts in radiology disagree on certain cases, particularly those with ambiguous features or poor image quality. Deep learning models, such as ResNet50 and DenseNet121, have shown strong performance in automated classification, but often struggle with uncertainty. When models fail to distinguish between confidently correct predictions and those requiring further review, they risk producing overconfident but incorrect diagnoses, which can negatively impact clinical decision-making. Addressing this limitation is essential for improving AI-assisted diagnostics and ensuring reliable deployment in medical settings.

This work investigates whether preprocessing techniques, specifically unsharp masking and histogram equalization, can improve classification reliability by enhancing image quality and reducing prediction uncertainty. Initially, we explored all 14 conditions available in the CheXpert dataset but refined our focus to the five competition tasks: Atelectasis, Cardiomegaly, Consolidation, Edema, and Pleural Effusion. This targeted approach enables more precise model tuning, improved classification metrics, and enhanced Grad-CAM visualizations. The input to our system consists of frontal

CXR images from CheXpert in both raw and preprocessed forms at three resolutions: 224×224, 384×384, and 512×512. We utilized pretrained ResNet50 and DenseNet121 models to classify these five conditions while integrating Grad-CAM to analyze model decision-making. By comparing models trained on raw and preprocessed images at different resolutions, we aim to determine whether preprocessing and resolution scaling improve interpretability and reduce uncertainty, ultimately contributing to more trustworthy AI-assisted diagnostics.

## II. RELATED WORK

Recent research in CXR classification has primarily focused on preprocessing techniques, image resolution effects, and model interpretability. Preprocessing is widely used to enhance image quality and improve deep learning model accuracy. Heidari et al. developed a preprocessing pipeline that incorporated diaphragm removal, bilateral filtering, and histogram equalization, which improved the classification accuracy of a VGG16-based convolutional neural network (CNN) by 6.5 percentage points [1]. Similarly, Sabottke and Spieler analyzed how image resolution affects deep learning models, finding that higher resolutions generally improve performance but reach a plateau beyond a certain threshold [2]. While these studies highlight preprocessing as a critical factor in model performance, they primarily evaluate accuracy rather than uncertainty or confidence estimation. Our study extends this research by analyzing whether preprocessing affects classification uncertainty and confidence across multiple image resolutions rather than focusing solely on accuracy.

Another challenge in CXR classification is model interpretability, as deep learning models often operate as black boxes in clinical applications. Grad-CAM (Gradient-weighted Class Activation Mapping) has been widely adopted to visualize which image regions contribute most to a model's decision. Özer et al. applied Grad-CAM to CXR classification models and found that low-quality images often generate misleading heatmaps, reducing clinician trust in AI-assisted diagnosis [3]. Similarly, Panwar et al. used Grad-CAM for COVID-19 detection, showing that model attention generally focused on infection-affected lung regions but sometimes misfocused on irrelevant structures [4]. While Grad-CAM improves explainability, it doesn't inherently quantify uncer-

tainty, a key limitation in medical AI. Whata et al. explored uncertainty quantification techniques in CXR classification, showing that uncertain predictions frequently occur in cases with overlapping disease features or low-resolution images [5]. The most widely used uncertainty quantification methods involve Bayesian Neural Networks (BNNs), which introduce stochasticity into model parameters to generate confidence estimates [6]. While BNNs are effective, they're computationally expensive and primarily quantify parameter-based uncertainty rather than dataset-related variability. Most deep learning models still require radiologists to manually assess uncertain predictions without a built-in confidence measure. Our study builds upon prior research by evaluating how preprocessing and resolution scaling influence classification uncertainty, ensuring that model confidence remains reliable in real-world applications.

### III. DATASET AND FEATURES

#### A. Dataset Description

The CheXpert dataset, developed by Stanford's Center for Artificial Intelligence in Medicine and Imaging (AIMI) [7], is a publicly available collection of 223,414 frontal and lateral CXRs from 64,540 patients. Diagnostic labels are extracted automatically from radiology reports using CheXbert, a natural language processing (NLP) model. This labeling method enables large-scale dataset creation but may introduce inconsistencies due to variations in radiology report phrasing.

To assess whether the model generalizes beyond CheXbert-generated labels, a radiologist-labeled validation and test set is incorporated for reference evaluation [10]. The validation set is labeled by a consensus of three board-certified radiologists, while the test set is labeled by five board-certified radiologists. These expert annotations serve as a comparative benchmark to determine whether classification and uncertainty estimates align with human assessments.

The dataset contains five diagnostic labels relevant to our study: Atelectasis, Cardiomegaly, Consolidation, Edema, and Pleural Effusion. These conditions correspond to the CheXpert competition tasks, allowing us to benchmark performance effectively.

#### B. Preprocessing and Image Resolution

To assess the impact of preprocessing and resolution scaling, we apply three different image resolutions: 224×224, 384×384, and 512×512 pixels. Images in the raw dataset are rescaled and resized but otherwise unaltered. For the preprocessed dataset, we apply:

- 1) **Unsharp Masking** to enhance edges and improve feature clarity.
- 2) **Histogram Equalization** to normalize brightness and contrast.

These preprocessing steps help determine whether improved image quality leads to better classification performance and reduced uncertainty. To reduce the computational overhead and training time, we applied these preprocessing steps up-front and saved the images in .jpg format with quality set to 95.

#### C. Predictive Task and Feature Extraction

The primary task is multi-label classification, where each image is classified into one or more of the five diagnostic categories. Instead of discarding uncertain labels, we retain them to study their effect on predictions. Additionally, we use Grad-CAM visualizations to analyze which image regions contribute most to model predictions, providing insight into decision-making processes. We extract deep features from the images using pretrained ResNet50 and DenseNet121 CNNs. No additional manual feature engineering (e.g., PCA or Fourier transforms) is performed, as CNNs autonomously learn hierarchical image representations.

1) *Example Images:* Figure 1 presents examples of CXRs from the dataset, illustrating the impact of resolution scaling and preprocessing. The top image represents the original X-ray, while the middle and bottom rows show the raw and enhanced versions at three different resolutions. The enhanced images demonstrate improved contrast and edge clarity, which may influence model performance in classification tasks.

The 224×224, 384×384, and 512×512 resolutions were selected to assess the trade-off between computational efficiency and feature preservation. Higher resolution images retain finer anatomical details that could improve classification accuracy for certain conditions. However, higher resolution images require significant computational resources to process, and the long training times often makes using them impractical.

Preprocessing techniques, including unsharp masking and histogram equalization, enhance image contrast and emphasize structural details such as lung boundaries and abnormalities. However, while these transformations may improve feature visibility, they also risk amplifying noise or introducing artifacts, which could affect the model's ability to generalize.

By comparing predictions on raw versus preprocessed images, we aim to determine whether preprocessing reduces uncertainty in model outputs. If enhanced images lead to more confident predictions or improved class separability,

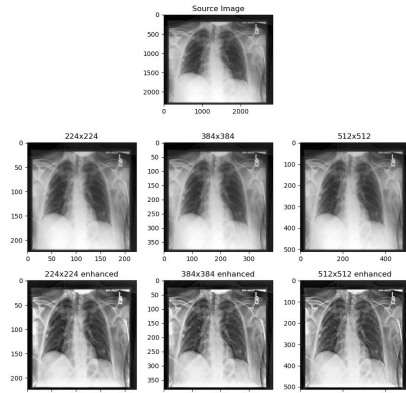


Fig. 1. **Example CXR at different resolutions and preprocessing levels.** The top image is the original CXR, the middle row shows raw images resized to 224×224, 384×384, and 512×512, and the bottom row presents the corresponding enhanced versions after unsharp masking and histogram equalization.

preprocessing could serve as an effective method for refining AI-assisted diagnosis.

#### IV. METHODS

##### A. Preprocessing the label data frames

Though the radiologist-labeled datasets provide an effective benchmark to compare our models to human performance, but they do not contain any uncertain labels. Instead, we reserved one batch of training images as the test set, then split the other two batches into training and validation sets using an 80:20 ratio. We then filter out all condition columns except the five competition task columns, and we retain only the images with frontal views to restrict the model scope. We mapped the N/A class values to 0, and the uncertain (-1) values to 2 to allow for the use of cross entropy loss. We performed these steps for all label data frames as necessary.

1) *Addressing Class Imbalances in the Training Set with Resampling:* Most classes are negative for most X-rays, leading to large imbalances between label distributions. However, certain classes are more affected than others, such as cardiomegaly, causing typical resampling techniques to fail. To mitigate this, we first remove rows from the training data frame that are negative for all five classes from the training set, then duplicate the data frame. We randomly remove 30,000 rows from one copy and resample 30,000 rows from the other with replacement. The samples are chosen using weights based on inverse class proportions to prioritize minority class labels, balancing the distributions more effectively as seen in Figure 2.

The final count of images in each dataset is as follows:

- **Training Set:** 92,742 images
- **Validation Set:** 30,215 images
- **Test Set:** 39,371 images
- **Radiologist-Labeled Validation Set:** 202 images
- **Radiologist-Labeled Test Set:** 518 images

##### B. Secondary Image Preprocessing Pipeline

To mitigate overfitting risks from image duplication due to resampling, we implement additional random preprocessing steps within the data loader. This pipeline consists of a random

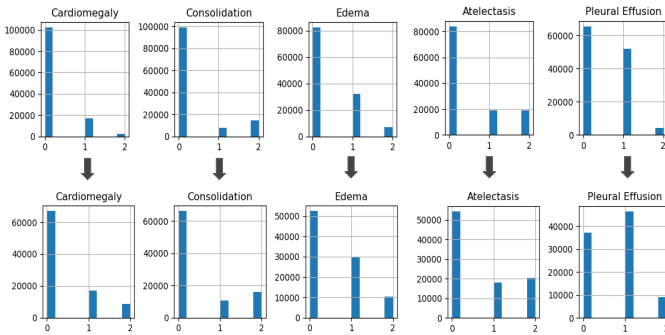


Fig. 2. **Visualizing the updated class distributions.** Although every class except Pleural Effusion remains dominated by negative instances, the class value proportions are more equal.

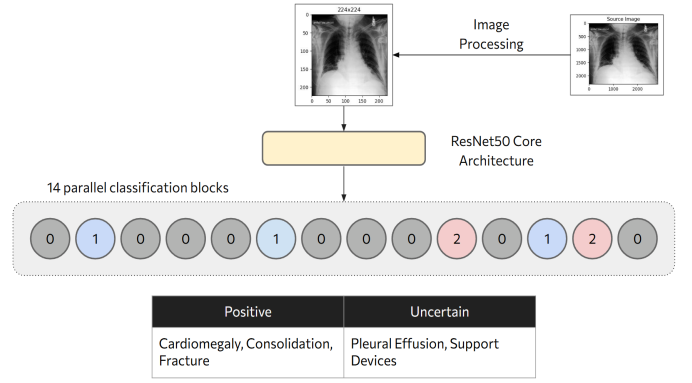


Fig. 3. **The customized pre-trained model template.**

This model configuration is designed to allow easy swapping between ResNet50 and DenseNet121 core architectures. It also allows for a separate cross entropy loss function to be applied to each output layer for specialized loss function applications

rotation by a 90-degree interval, another random rotation by a 30-degree interval, and probabilistic center-cropping at different magnitudes based on a normal distribution. No additional transformations were applied, both to minimize introduced bias from resampling images with replacement, and to isolate the effect of our preprocessing steps.

##### C. Model Configurations

For our prediction tasks, we use pre-trained ResNet50 and DenseNet121 CNN models as the core architecture. ResNet50 is a commonly used model structure for image classification tasks which utilizes skip connections to better propagate information through the layers, allowing for the use of deep networks [8]. DenseNet121 similarly utilizes skip connections, except that all layer outputs are interconnected, resulting in a more compact model representation that allows for efficient feature reuse [9]. We used the model structures and default weights stored in the torchvision library, both of which were pre-trained on the ImageNet dataset.

We built twelve models, one for each combination of model architecture (ResNet50 and DenseNet121), image resolution (224x224, 384x384, and 512x512), and image preprocessing pipeline (raw and enhanced). We will refer to the models with the following shortened format in this paper: ResNet-224 and ResNet-224-enhanced.

We used pre-trained ResNet and DenseNet models as the foundational models for transfer learning, but our multi-label classification problem required additional complexity. To customize them for our task, we replace the final layer with five parallel output layers, one for each condition, with three output weights each, one for each class value. This configuration allows for easy swapping of core model architectures, and it allows us to visualize the model's decision-making for all conditions simultaneously.

We implement a separate weighted cross-entropy loss function for each class based on individual class label distributions. The loss for each epoch is defined as the average loss across

TABLE I  
BREAKDOWN OF MODEL LEARNING RATES BY CONFIGURATION

| Image Size | Model Configuration | $lr1$  | $lr1$  |
|------------|---------------------|--------|--------|
| 224x224    | ResNet50            | 1e-3   | 1e-6   |
| 224x224    | DenseNet121         | 1e-3   | 1e-6   |
| 384x384    | ResNet50            | 2.5e-3 | 2.5e-6 |
| 384x384    | DenseNet121         | 2.5e-3 | 2.5e-6 |
| 512x512    | ResNet50            | 2.5e-3 | 2.5e-6 |
| 512x512    | DenseNet121         | 2.5e-3 | 1e-7   |

each classification layer. We employed the Adam optimizer with an initial learning rate ( $lr1$ ) for the first five epochs and a secondary smaller learning rate ( $lr2$ ) for the remaining epochs. The breakdown of learning rates for each model configuration is listed in table 1. All layers except for the final classification layers are frozen through the entire training process to mitigate overfitting. We trained each model for a minimum of 5 epochs and a maximum of 50 epochs with early stopping implemented when the difference in validation loss is less than 0.0005.

Running these models efficiently required significant parallelization efforts. We split the modeling tasks between two computers, both with NVIDIA GeForce RTX 3060 GPUs and AMD Ryzen 5 7600X 6-core CPUs. One computer has 64 GB of memory and the other has 32 GB of memory. For each model, we utilize a training image data loader class with a maximum batch size of 32 which randomizes the secondary image processing pipeline upon object instantiation. To allow for parallel image processing, the data loaders were instantiated with 4 persistent workers. The training images are additionally shuffled to avoid overfitting to a specific image order.

#### D. Evaluation Metrics

As our primary goal is to explore how various preprocessing techniques may affect classification tasks, we explore three sets of metrics to evaluate model performance:

- 1) **Classification Metrics:** A focused analysis on how well the model classifies the presence of the five conditions.
- 2) **Uncertainty Analysis:** A focused analysis on how well the model correctly classifies certain and uncertain instances.
- 3) **Grad-CAM Analysis:** A focused analysis on the average similarity between output Grad-CAM heat map representations.

Medical diagnosis as a machine learning task can have dire consequences if the model makes incorrect predictions - therefore, we use precision and recall as our main classification metrics. We also use the F1 score which, as the harmonic mean of precision and recall, measures how well the model balances both precision and recall. As the class distributions are highly skewed, we use the balanced error rate (BER) which averages the error between both positive and negative classes.

We use precision and recall to evaluate how well the model distinguishes between certain and uncertain instances. By examining the proportions of certain instances being classified

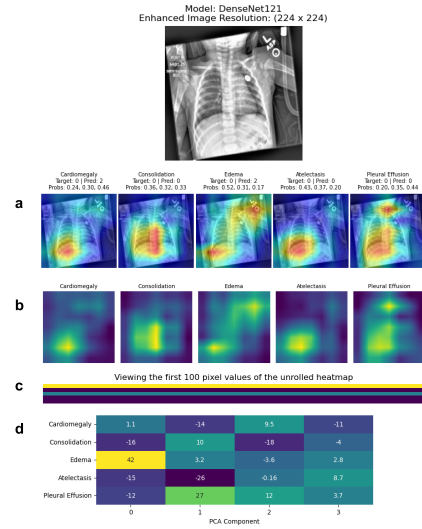


Fig. 4. **Visualizing the Heat Map PCA Analysis.** (a) Viewing the heat maps overlaid onto the original CXR (b) Extracting the individual heat maps (c) Unrolling the heatmaps using a Hilbert curve index and combining the 1D vectors into a 2D matrix format (d) Reducing the dimensionality to just 4 dimensions via PCA

as uncertain and vice versa, we can determine which model configuration leads to the most reliable results, as well as which medical conditions are harder to classify with a high degree of certainty.

We also utilize Grad-CAM to analyze heatmaps of regions important for decision-making between conditions to determine sources of confusion within the images. To achieve this, we took a sample of 500 CXRs from the test set, and for each CXR, we extracted the Grad-CAM heatmaps for each condition. Figure 4 shows an example of a set of heatmaps produced for a specific model configuration. For each sample CXR, we unrolled the five Grad-CAM heatmaps into 1-D representations using a Hilbert curve index to retain spatial relationships, and we combined the resulting vectors into a matrix. Next, we reduced each matrix to five components using a separate PCA applied to each set of heatmaps. We then aggregated condition-wise dot product similarity metrics across the entire sample for each model configuration - this metric encodes the average similarity in Grad-CAM heatmap representations between conditions on the same CXR. Highly similar heatmaps between conditions may indicate overfitting to specific features. Finally, we subtracted the average dot product similarity matrices from the enhanced models from the average dot product similarity matrices from the raw models - this metric encodes how image enhancements may affect the similarity between heatmap representations. Large positive differences in dot product similarity indicate that the preprocessing methods lead to more unique condition-wise heatmap representations on average, while large negative differences indicate an increase in condition-wise similarity.

## V. RESULTS

### A. Classification Performance Across Conditions

Our primary evaluation focuses on how well different model configurations classify the presence of the five conditions. Since medical diagnosis requires a balance between minimizing false positives and false negatives, we use F1 Score as the primary metric and BER instead of accuracy, which averages the error rate across both classes to mitigate the effects of class imbalance.

For a given class, F1 Score is defined as:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

where Precision measures the proportion of true positive predictions among all positive predictions, and Recall measures the proportion of actual positives that were correctly identified.

Since class distributions are skewed, BER is used to evaluate classification balance across both classes:

$$BER = \frac{1}{2} \left( \frac{FP}{FP + TN} + \frac{FN}{FN + TP} \right) \quad (2)$$

where FP and FN represent false positives and false negatives, respectively.

1) *Model Generalization to Radiologist-Labeled Data:* To assess model generalization beyond CheXpert-labeled training data, we compare performance against the radiologist-labeled test set, which serves as a stronger clinical ground truth. Figure 5a presents the BER difference (Radiologist - CheXpert), where negative values indicate better alignment with expert labels, while positive values suggest potential bias toward CheXpert’s automated labels.

Most models exhibit stable or slightly improved BER on the radiologist-labeled test set, suggesting limited overfitting to CheXpert’s labels. Pleural Effusion and Consolidation show consistently negative BER differences, indicating better alignment with expert annotations. Conversely, Atelectasis shows increased BER in multiple models, potentially due to inconsistencies between CheXpert and expert annotations. Cardiomegaly and Edema exhibit minor BER shifts, suggesting performance stability across labeling sources.

Evaluating models against radiologist-labeled data provides insight into clinically meaningful generalization, ensuring models capture relevant diagnostic features rather than dataset-specific biases.

2) *Classification Performance and Best Model Selection:* Figure 5b compares F1 Scores across models and conditions, showing variation between architectures. Base models generally outperform enhanced versions, suggesting preprocessing doesn’t consistently improve classification performance. Pleural Effusion and Edema achieve the highest F1 Scores, while Consolidation and Atelectasis remain difficult to classify, likely due to subtle radiographic features and label uncertainty.

Model performance also varies by architecture. DenseNet models perform well for Edema and Cardiomegaly, while ResNet models achieve stronger performance for Pleural Effusion. Larger models (ResNet-512, DenseNet-512) don’t consistently outperform smaller architectures like DenseNet-224, suggesting that higher resolution doesn’t always improve classification performance.

To identify top-performing models, Figure 6a summarizes the best models per condition based on F1 Score and BER, while Figure 6b highlights Precision and Recall leaders. Figure 6c presents the frequency of best model selections across conditions.

DenseNet-224 appears most frequently as the best-performing model, suggesting strong generalization and efficiency. ResNet-512 excels in specific cases, particularly Pleural Effusion, highlighting a trade-off between computational cost and classification performance.

### B. Uncertainty Analysis

As seen in Figure 7, all models performed relatively well with respect to precision, though notably, all models performed better in identifying uncertain instances for Pleural Effusion and Cardiomegaly. In particular, the ResNet-512 models achieved the highest precision across all model configurations for Cardiomegaly. However, with respect to recall, model performance varied greatly. This effect is most evident with Pleural Effusion, though Cardiomegaly also has a significant drop in recall with the ResNet-512 models.

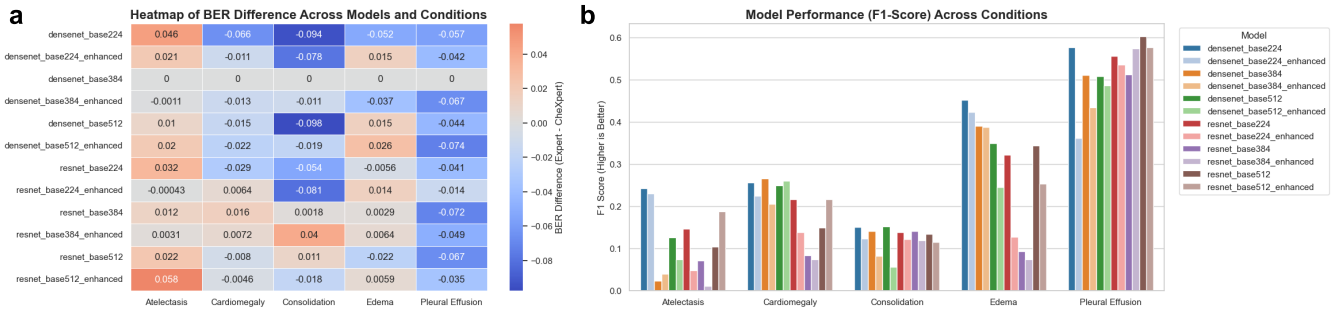


Fig. 5. **Balanced Error Rate (BER) Difference and F1 Score Across Conditions.** (a) BER difference (Radiologist - CheXpert) across models and conditions. Lower (negative) values indicate that the model generalizes well to expert-labeled data, while higher values suggest potential bias toward CheXpert labels. (b) F1 Score across models and conditions. Higher F1 scores indicate better classification performance in distinguishing positive vs. negative cases.



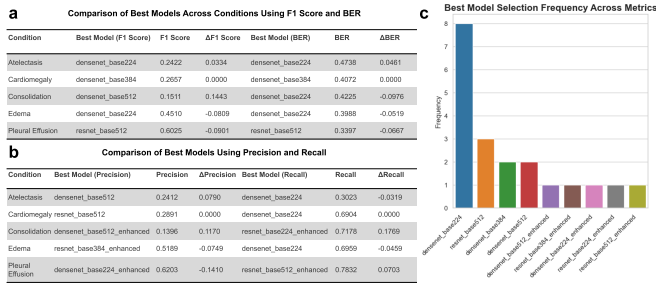


Fig. 6. **Comparison of Best Models Across Metrics and Selection Frequency.** (a) Best models selected per condition based on F1 Score and BER. Lower BER and higher F1 scores indicate better classification performance. (b) Best models selected based on Precision and Recall. Higher precision minimizes false positives, while higher recall minimizes false negatives. (c) Frequency of model selection across all evaluation metrics, showing which models were most frequently identified as the best across conditions.

The larger models had better recall metrics for Atelectasis and Consolidation, suggesting that these models benefited from the increased level of detail. The enhanced variations of ResNet-224 and DenseNet-512 achieved higher metric with Edema than the raw variations. Similarly, the enhanced variation of ResNet-224 performed better than its raw counterpart with Cardiomegaly. However, these appear to be in the minority, as most other model pairings either had minimal difference, or the preprocessed variations performed worse.

Although all models displayed sharp precision-recall trade-offs, the results were fairly stable across all models for Edema, where both precision and recall remained at or above 0.7. As both precision and recall are fairly important metrics

All models had trouble correctly classifying uncertain instances, though the trend differs based on whether the uncertain instance was classified as certain positive, or certain negative. Pleural Effusion consistently had the worse performance across all models, with fairly high rates of uncertain instances being classified as either positive or negative. All models also performed poorly with Edema, especially with the larger models classifying large proportions of uncertain instances as negative. All models often misclassified uncertain instances for Cardiomegaly as negative, though the worst-performing models in this category were the ResNet-512 pairing.

All models tended to misclassify positive instances as uncertain for both Atelectasis and Edema. All models also tended to misclassify negative instances as uncertain across all models, with the notable exception of Pleural Effusion. Again, the larger models tended to perform worse, particularly with Atelectasis and Consolidation, as for both conditions, the larger models misclassified half or more of all negative instances as uncertain.

Between all conditions, the models were most reliably able to distinguish between certain and uncertain images for Consolidation, especially with respect to the positive instances. Negative instances were the most likely to be confused across all models - as most conditions are negative across all images, this may be due to the model learning spurious patterns and overfitting to the training set.

The PCA analysis across the condition-wise Grad-CAM representations of 500 sampled CXRs revealed a large amount of variance displayed in the ResNet models as can be seen in Figure 10. All DenseNets achieved more than 80% variance explained with just two principal components while many ResNets needed at least four components to achieve similar FVE.

Figure 8 shows the average dot product similarity between Grad-CAM heat maps of different conditions across all models. Aggregated across a sample of 500 CXRs, this metric allows us to estimate how often models create unique Grad-CAM heat maps for different conditions. While unique Grad-CAM maps may pinpoint diagnoses more effectively, they may also indicate overfitting, especially to noise present in the image.

Across all model configurations, most conditions manifested unique heat maps as can be observed by the large number of negative values. In particular, Atelectasis and Edema heat maps had an average dot product similarity of about -1075. However, Edema and Pleural Effusion seem to manifest similar heat maps, with an average dot product similarity of about 335. Considering both conditions concern excess fluid buildup in or around the lungs, this similarity makes sense. However, similar Grad-CAM heat maps may confuse the model and could lead to less generalizability.

Figure [9] shows the difference in average dot product similarity between the models trained on preprocessed images and the models trained on the raw images. Positive values indicate that preprocessing lead to more unique maps, while negative values indicate that preprocessing lead to more similar maps. As can be seen, preprocessing the images improved the uniqueness of Grad-CAMs across all model configurations, though the effect is not uniform and is quite noisy. The DenseNets had the largest differences, indicating that preprocessing had a more significant effect on these models.

Preprocessing appeared to have the most significant effect on the DenseNet-224 models - after preprocessing, Consolidation's heat maps became extremely similar to all other conditions except for Cardiomegaly. On the other hand, preprocessing appeared to increase the similarity between the Edema, Atelectasis, and Pleural Effusion grouping. However, none of these results appear to correlate with an increase of performance in our earlier analysis - in fact, many of the models trained on enhanced performed much worse than their raw counterparts. This indicates that these preprocessing methods may not have been effective for this task.

While aggregated metrics across a sample of Grad-CAM heat maps did not offer much insight, analyzing heat maps for an individual CXR prove to be much more useful. As can be seen in figure 11, the ResNet50 models began overfitting to noise as the image resolution increases. This effect is also observed in the DenseNet models, albeit to a lesser degree. For this particular CXR, the image enhancements appears to expand the region of interest for the model for the 224x224 models, but for the larger models, the effect is a bit more unclear.

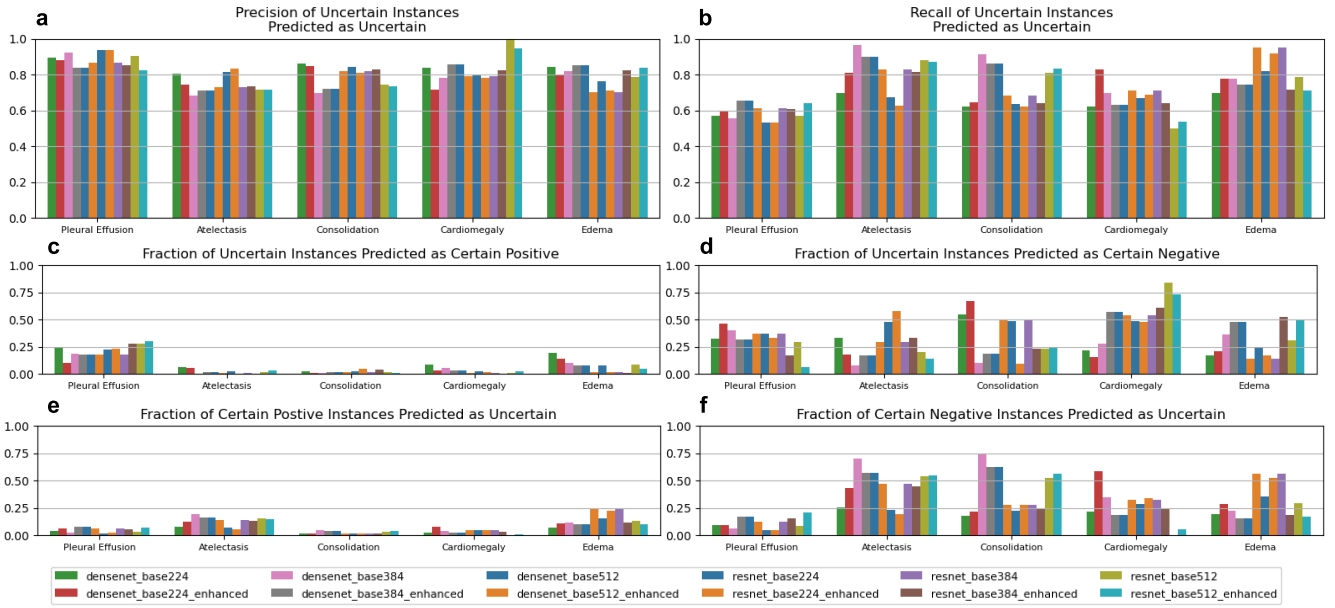


Fig. 7. **Comparing Model Performance Based on the Uncertain Label.** These plots examine how well the models distinguish certain images from uncertain images based on the presence of the uncertain label (2). The top plots focus on precision and recall, while the lower plots focus on proportions of misclassifications..

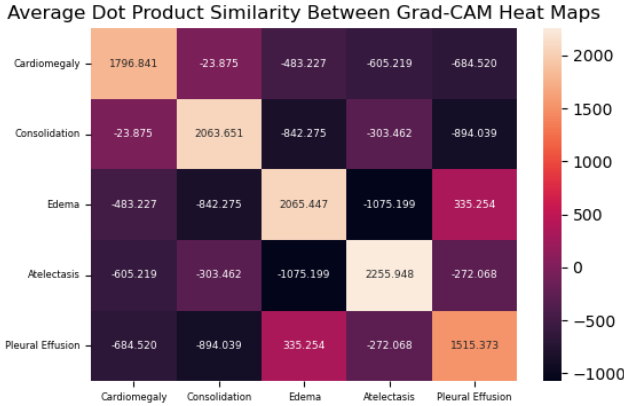


Fig. 8. **Visualizing the average dot product similarity between Grad-CAM heat maps across all models.** For most models, the output Grad-CAM heat maps were fairly unique, with only Pleural Effusion and Edema producing heat maps with a positive average similarity. Consolidation and Cardiomegaly were the least correlated, whereas Atelectasis and Edema often had very different heat maps.

These results seem to suggest that smaller models are effective enough for CXR-related tasks, as they are able to identify relevant regions while being much more robust against overfitting to noise. They also suggest that DenseNets may perform better, as the ResNet heat maps had more variance, especially many of the larger model variations.

## VI. CONCLUSION

This study evaluated the effects of preprocessing, image resolution, and uncertainty quantification on deep learning-based CXR classification. Using pre-trained ResNet50 and

DenseNet121 CNNs, we compared models trained on raw and preprocessed images at resolutions of 224×224, 384×384, and 512×512 pixels. DenseNet121 at 224×224 resolution achieved the best overall performance, balancing accuracy, computational efficiency, and interpretability. Higher-resolution models, particularly ResNet50 at 512×512, tended to overfit, as evidenced by Grad-CAM visualizations highlighting irrelevant image regions. Preprocessing techniques—unsharp masking and histogram equalization—improved visual clarity but didn’t consistently improve classification accuracy, instead introducing variability in model confidence. Conditions such as Atelectasis and Edema remained challenging to classify accurately, displaying consistently higher uncertainty regardless of preprocessing.

Our original goal was to reduce classification uncertainty and improve model interpretability through preprocessing and resolution adjustments. While we successfully identified a computationally efficient model configuration (DenseNet-224) that balanced accuracy and uncertainty, preprocessing yielded mixed results. CheXlocalize evaluation revealed a generalization gap between NLP-extracted labels (CheXbert) and radiologist annotations, suggesting inherent limitations when relying exclusively on NLP-derived datasets. Grad-CAM visualizations provided meaningful insight into model decision-making but proved less reliable in uncertain predictions, highlighting areas for further improvement.

Future work should explore advanced uncertainty estimation methods, such as Bayesian neural networks or Monte Carlo dropout, to improve reliability in challenging diagnostic cases. Additionally, alternative preprocessing techniques or the integration of clinical metadata could further enhance the capture

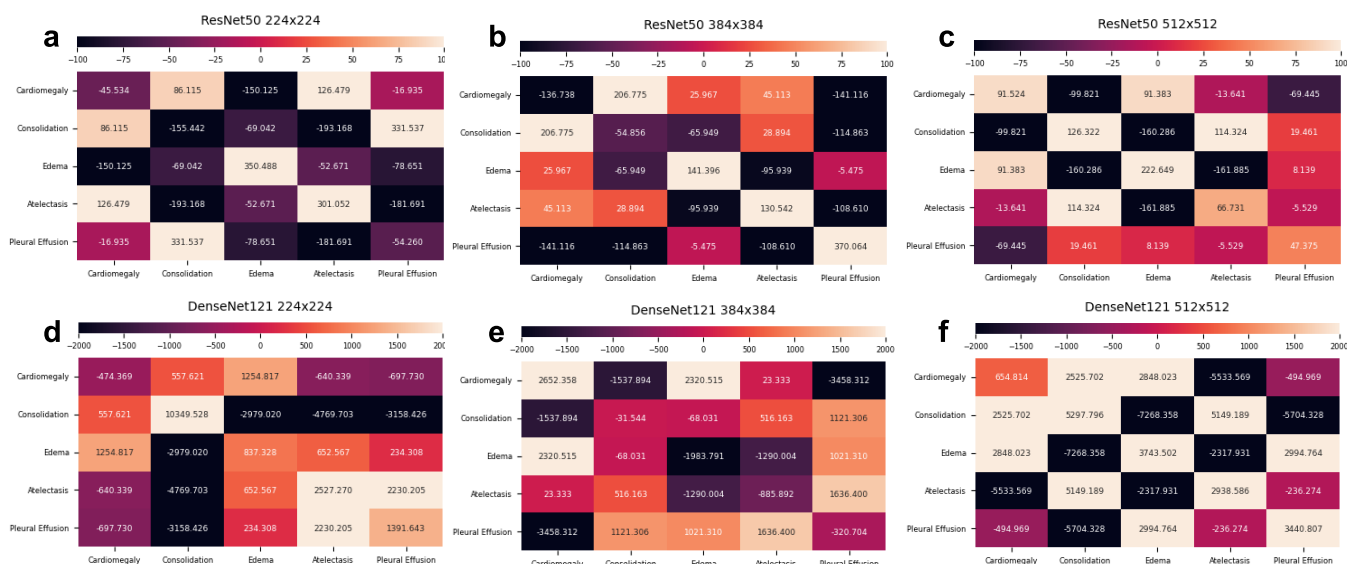


Fig. 9. Visualizing the difference in average dot product similarity between Grad-CAM heat maps of different conditions. Training image preprocessing had the largest effect on the DenseNet models (9.d-9.f) while the results were much noisier for the other models.

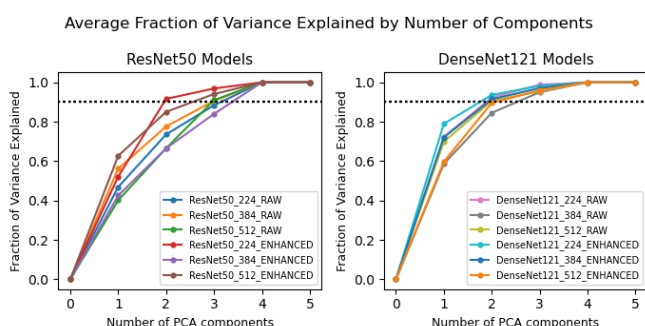


Fig. 10. Visualizing the Fraction of Variance Explained. The DenseNet models had steeper curves, indicating that the PCA components explained the variation in the data quite well compared to the ResNet models.

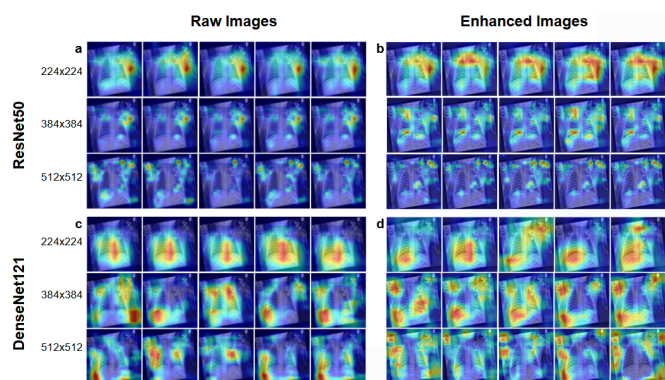


Fig. 11. Visualizing the Fraction of Variance Explained. The DenseNet models had steeper curves, indicating that the PCA components explained the variation in the data quite well compared to the ResNet models.

of diagnostically relevant features. Evaluating models across external and diverse datasets will also be crucial to confirm the generalizability and clinical utility of these approaches. Different model architectures should also be explored - there are many variations of ResNet and DenseNet structures, and there are other model architectures commonly used in this field such as ViT. Quantifying the uncertainty and explainability of these models provides useful insight that can help guide future work on image classification tasks.

## CONTRIBUTIONS

John formulated and finalized the data pipeline used to preprocess and prepare the CXR images for neural network training. He performed exploratory data analysis (EDA) focused specifically on image characteristics, trained and fine-tuned the ResNet50 and DenseNet121 models, and significantly contributed to developing and finalizing the Grad-CAM visualization pipeline.

Zoey conducted exploratory data analysis emphasizing patient demographics and class distributions, informing methodological decisions and interpretation of results. She performed analyses on classification metrics and model generalization, critically evaluating the models' performance and robustness. Zoey also designed and implemented the project website, incorporating interactive visualizations to effectively present key findings and analytical outcomes.

Both team members collaborated equally on the milestone reports, literature review, overall conceptual framework, and interpretation of results.

## REFLECTIONS

If we had the opportunity to redo this project, we would reconsider several methodological decisions to better address the limitations we encountered. A significant challenge was



the absence of explicit uncertainty labels in the radiologist-labeled dataset (CheXlocalize), forcing us to rely heavily on the CheXpert training dataset, which included uncertain NLP-derived labels from CheXbert. As a practical alternative, we had to create internal splits from the training dataset to evaluate uncertainty quantification. Ideally, we could have supplemented our data with other publicly available datasets explicitly annotated for uncertainty or inter-rater variability (e.g., subsets of NIH ChestX-ray14 or MIMIC-CXR). Even limited use of such datasets might have provided stronger validation and clearer insights into our models' uncertainty handling. We were also limited by computational constraints, so we were unable to refine the models to the best of our ability within a reasonable time frame. Despite many efforts to finetune the process, many of the larger models took multiple hours to train, so hyperparameter tuning was difficult. With more time and computational power, we could have tested more preprocessing methods and model architectures to gain a more complete understanding of the sources of uncertainty in the dataset and model.

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## REPLY TO REVIEW

One review focused on how feature engineering could improve the model training times. Originally, we planned to keep all 14 condition columns in the dataset - however, we were quickly faced with extremely long model training times. For example, the 512x512 image datasets were forecasted to take over 2 entire days to train. One review noted how feature engineering prior to training to adjust our task slightly could improve our results. They suggested that we could combine conditions to predict them in groups, and to decrease our sample sizes. This review led us to reevaluate our task, and we settled on only retaining the five in our analysis. These are the CheXpert competition tasks, so there is precedence to using only these columns. We also redeveloped our resampling technique in order to reduce the amount of images being analyzed.

Another review noted how we originally only used 8 epochs to train the model, and though this was originally done to reduce computational requirements, we found that our optimizations were enough to allow for more training epochs.